

Graph-Based Movie Recommendation Using Link Prediction on MovieLens-100K

Bushra Sadaf

*Dept. of Computer Engineering
Habib University
Karachi, Pakistan
bs09267@st.habib.edu.pk*

Anushah Rizvi

*Dept. of Communication and Design
Habib University
Karachi, Pakistan
ar09232@st.habib.edu.pk*

Abstract—This paper presents a graph-based movie recommendation system built on the MovieLens-100K dataset using Neo4j and the Graph Data Science (GDS) library. The recommendation problem is framed as link prediction on a bipartite User-Movie graph, where existing ratings serve as known edges and the objective is to predict unobserved interactions [3]. Four feature engineering strategies are systematically evaluated: degree-based features, degree combined with rating statistics, hybrid embeddings with structural and node features, and embedding-only representations using FastRP. Each configuration is trained using logistic regression, and performance is evaluated using Area Under the Precision-Recall Curve (AUCPR). A random forest classifier is additionally applied to embedding-based features to assess non-linear gains. Results show that learned graph embeddings significantly outperform handcrafted features. The system concludes by generating personalized recommendations for users based on predicted link probabilities.

I. INTRODUCTION

Recommender systems play a critical role in modern digital platforms by enabling personalized content delivery in domains such as entertainment, e-commerce, and social media. Traditional approaches such as collaborative filtering and matrix factorization rely heavily on user-item interaction matrices and often struggle with sparsity and cold-start limitations, where new users or items lack sufficient interaction history [1].

Graph-based recommender systems provide an alternative by modeling users and items as nodes and interactions as edges, enabling richer structural learning. Neo4j and graph machine learning frameworks allow direct representation and analysis of the structural properties of graphs.

The MovieLens-100K dataset contains 100,000 ratings from 943 users and 1,682 movies [2]. In this work, we construct a bipartite graph where users are connected to movies via RATED relationships. The recommendation task is formulated as link prediction: estimating the likelihood of unseen user-movie interactions.

We evaluate four feature configurations ranging from simple structural statistics to learned embeddings. The goal is to analyze how increasing representational complexity impacts predictive performance under AUCPR, which is suitable for highly imbalanced link prediction problems [5].

II. METHODOLOGY

A. Data Cleaning & Graph Construction

The dataset was cleaned and structured into a graph using Neo4j. Duplicate prevention was ensured through uniqueness constraints on `userId` and `movieId`. Movie genre attributes, originally encoded as binary flags, were consolidated into a single pipe-separated string to reduce graph complexity while preserving semantic information. This design choice avoids unnecessary node explosion while maintaining full feature details.

B. Dataset and Preprocessing

The dataset includes user demographic attributes (age, gender, occupation), movie metadata (title, release year, IMDb URL), and explicit ratings on a scale of 1–5. Timestamps are included to preserve interaction ordering. These components are mapped into node and edge properties for graph construction.

C. Graph Model and Data Loading

The graph consists of three main components: User nodes, Movie nodes, and RATED relationships. User nodes store behavioral attributes, while movie nodes store metadata and genre information. Each RATED relationship stores a rating value and timestamp. The final graph contains 943 users, 1,682 movies, and 100,000 edges, forming a sparse bipartite structure.

D. Graph Statistics

Structural analysis shows a sparsity level of approximately 6.3%, which is typical for recommendation datasets. Users have an average of 106 ratings, while movies receive 59 ratings on average. The most popular movie is Star Wars (1977), while User 405 is the most active rater. No isolated nodes exist, indicating full connectivity across the interaction graph.

E. Graph Analytics

Exploratory analysis reveals strong behavioral patterns in the dataset. Drama and Comedy dominate user engagement, indicating skewed genre preferences. User overlap analysis identifies strong similarity between users based on shared

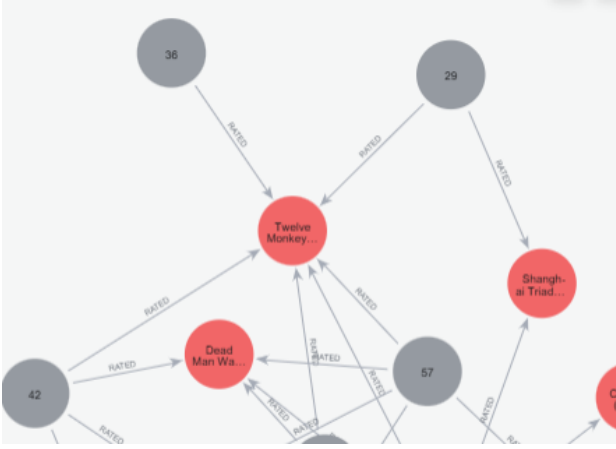


Fig. 1. Neo4j bipartite User-Movie graph structure for MovieLens-100K dataset

ratings. A 2-hop traversal (User $\rightarrow$ Movie $\rightarrow$ User $\rightarrow$ Movie) demonstrates the presence of indirect relationships that can be leveraged for recommendations. Degree centrality analysis shows that a small number of movies act as hubs, reflecting central and highly popular nodes commonly observed in real-world recommender systems.

F. Node Feature Computation

Four primary features are computed for model training:

- User.degree: number of rated movies (activity level)
- Movie.degree: number of user interactions (popularity)
- User.avgRating: average rating behavior of user
- Movie.avgRating: perceived quality of movie

G. Model Implementation & Training

The GDS Link Prediction pipeline is used for all experiments. The graph is projected into memory, and HADAMARD feature combiners generate edge-level features. The dataset is split using an 80/20 train-test ratio with cross-validation. Logistic regression is used as the base classifier, while random forest is used for embedding evaluation. FastRP embeddings (64 dimensions) are generated using Neo4j’s Graph Data Science (GDS) library [7].

H. Experimental Configurations

Four configurations are evaluated:

- Exp. 1: Degree-only structural features
- Exp. 2: Degree + average rating features
- Exp. 3: Degree + rating + FastRP embeddings
- Exp. 4: Embedding-only (FastRP + Random Forest). Tests whether learned structural representations complement handcrafted features [4].

III. RESULTS AND DISCUSSION

A. Evaluation & Analysis

Table I presents AUCPR results across all experiments.

TABLE I
AUCPR RESULTS ACROSS EXPERIMENTS

Experiment	AUCPR	Classifier
Degree Only	0.3230	Logistic Regression
Degree + AvgRating	0.3230	Logistic Regression
Degree + AvgRating + Embedding	0.3230	Logistic Regression
Embedding Only (LR)	0.8068	Logistic Regression
Embedding Only (RF)	0.9010	Random Forest
Traditional ML Baseline	0.6800	Random Forest

1) *Performance Analysis:* The degree-only model achieves low predictive performance (AUCPR 0.3230), indicating that simple structural statistics are insufficient for modeling user-movie interactions. Adding rating-based features does not significantly improve performance, suggesting limited linear separability in handcrafted features.

In contrast, embedding-based models show a major improvement, achieving AUCPR 0.8068. This demonstrates that FastRP embeddings effectively capture multi-hop structural dependencies in the graph. The best performance (AUCPR 0.901) is achieved using random forest on embeddings, indicating that non-linear decision models better use embedding’s structure.

2) *Recommendation Output:* The best-performing model generates top-10 recommendations by filtering unseen movies and ranking predicted probabilities. For User 64, recommendations are dominated by highly connected movies, reflecting strong influence of node popularity and graph connectivity patterns. While this improves accuracy, it introduces a bias toward frequently rated items, reducing recommendation diversity.

B. Reproducibility

All preprocessing steps, Cypher queries, and GDS pipelines are documented in a structured GitHub repository [8]. The repository includes separate modules for data ingestion, feature generation, and model training to ensure reproducibility.

IV. CONCLUSION AND FUTURE WORK

This paper presented a graph-based movie recommendation system using Neo4j, Cypher, and the Graph Data Science (GDS) Link Prediction framework on the MovieLens-100K dataset. The recommendation task was formulated as a link prediction problem on a bipartite User-Movie graph, where existing ratings represented known interactions and predicted links represented potential recommendations. Experimental results show that FastRP embeddings significantly outperform handcrafted features, with the best performance achieved using random forest on embeddings (AUCPR 0.901).

These results demonstrate that learned graph embeddings capture substantially richer graph structural properties than degree-based or rating-based node statistics. They also indicate that model performance is strongly dependent on the quality of feature representations, with embedding-based approaches consistently enabling stronger predictive performance across classifiers.

Overall, this study highlights the effectiveness of graph representation learning for recommendation tasks and confirms

that structural embeddings provide a significant advantage over handcrafted features in sparse bipartite interaction graphs.

Several directions remain for future work. First, richer node features, such as genre embeddings, or enhanced user demographics, could be incorporated to improve representation quality. Second, more expressive graph representation techniques such as Node2Vec or GraphSAGE may capture higher-order and non-linear structural relationships that FastRP may not fully model [6]. Third, temporal modelling of user rating behaviour could be introduced to account for evolving preferences over time, improving personalization accuracy. Finally, incorporating diversity-aware ranking strategies along with online or simulated user evaluation mechanisms (e.g., modelling actual user click or interaction behaviour in an online environment) would provide a more realistic assessment of recommendation quality beyond offline AUCPR evaluation.

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